Red Hat Ceph Storage

# Release Notes

9.1

IBM

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# Release notes for 9.1

Red Hat Ceph Storage is a hardened, qualified, secure, and supported enterprise software curated from the Ceph open-source project and delivered by Red Hat.

## New features and enhancements

This section lists all the major updates, and enhancements introduced in this release of Red Hat Ceph Storage.

### `cephadm` utility

**Improved `cephadm` upgrade workflow with failure-domain scoped OSD upgrades**

Previously, OSD upgrades were not aligned with physical failure domains, leading to less controlled upgrade operations.

With this enhancement, `cephadm` supports failure-domain scoped upgrades using CRUSH bucket parameters, enabling controlled, staged upgrades within racks, chassis, or hosts. This improves operational safety and upgrade efficiency.

(IBMCEPH-11980)

**Added automated deployment of Ceph Object Gateway D3N/D4N using `cephadm`**

Previously, deploying Ceph Object Gateway D3N/D4N configurations required manual setup and orchestration, increasing complexity and risk of misconfiguration.

With this enhancement, `cephadm` now automates the deployment of Ceph Object Gateway D3N/D4N configurations, simplifying setup and ensuring consistent deployment across clusters. This reduces operational effort and improves reliability for object storage workloads.

(ISCE-773)

**Image vendor detection enhancement in cephadm inspect-image**

Previously, there was no reliable way to determine whether a Ceph container image was built by IBM or Red Hat when inspecting images at runtime.

With this enhancement, `cephadm inspect-image` now detects and reports the image vendor as either `ibm` or `redhat`. This detection is based on the `/etc/redhat-storage-release` file inside the container image, enabling `cephadm` to identify the vendor from the release string. As a result, this enhancement provides clearer visibility into container image origin, improving troubleshooting, validation, and operational awareness when working with Ceph images.

(IBMCEPH-13854)

**SSH hardening and command execution enhancements for `cephadm`**

Previously, configuring a `cephadm` user required manual steps, including copying SSH public keys and updating the sudoers file. Additionally, `cephadm` executed commands directly over SSH without a centralized control mechanism. Arbitrary command execution paths were not standardized, and SSH hardening was not available.

With this enhancement, new SSH hardening capabilities and supporting commands are introduced:

- Automated SSH user setup.

```
cephadm setup-ssh-user --ssh-user SSH_USER --ssh-pub-key SSH_PUB_KEY
```

  The `cephadm setup-ssh-user` command copies the SSH public key and updates the sudoers configuration. The existing `cephadm set-user` command now invokes this workflow automatically to ensure the user is correctly configured.

- Standardized command execution interface.

```
cephadm exec --command COMMAND
```

All bash command execution in `cephadm` now uses this interface, providing consistent handling across the `mgr/cephadm` code.

– SSH hardening enablement.

```
ceph cephadm prepare-host-and-enable-ssh-hardening USER [HOST_LABEL]
```

This command prepares the host, configures the `cephadm` user, installs the required `cephadm` version if needed, and enforces command execution through the `cephadm_invoker.py` script. As a result, `cephadm` provides a more secure and consistent SSH execution model, reduces manual configuration steps, and enables hardened command execution paths across managed hosts.

(ISCE-404, IBMCEPH-13315)

## Ceph Dashboard

**New support for `cephadm`-managed SSL certificates for Ceph Object Gateway services**

Previously, SSL certificates required manual configuration and were not integrated with `cephadm`.

With this enhancement, `cephadm`-managed SSL certificates are now supported for Ceph Object Gateway deployment through the dashboard, enabling automated certificate handling and simplifying secure service configuration.

(IBMCEPH-13890)

**Enhanced MDS status monitoring**

Previously, monitoring the health and state of Metadata Servers (MDS) provided limited visibility into detailed status and operational metrics.

With this enhancement, the dashboard provides improved MDS status monitoring, enabling better visibility into performance, health conditions, and operational state. This enhances troubleshooting and cluster observability.

(ISCE-3031)

**Added CephFS subvolume metrics to dashboard**

Previously, subvolume-level metrics were not readily available in the dashboard, limiting visibility into usage and performance.

With this enhancement, CephFS subvolume metrics are exposed in the dashboard, enabling administrators to monitor usage patterns and resource consumption more effectively.

(ISCE-3111)

**Certificate Manager (`CertMgr`) support for Ceph Dashboard**

The dashboard now supports `CertMgr` integration with `cephadm`-signed certificates, allowing certificates to be automatically generated and managed by `cephadm`. This simplifies certificate lifecycle management and reduces manual administration.

The dashboard also monitors certificate health and displays warnings for certificates nearing expiration and alerts for certificate-related failures or misconfigurations. This helps administrators proactively identify and resolve certificate issues, ensuring secure and uninterrupted cluster operations.

(ISCE-2442, ISCE-1407)

**New Ceph Object Gateway multisite archive zone support**

Previously, the Ceph Dashboard supported configuration of standard Ceph Object Gateway multisite replication but did not support creating or managing archive zones. As a result, users had to rely on CLI commands to configure archive zones, leading to an inconsistent user experience and more complex day-two operations.

With this enhancement, the Ceph Dashboard now supports creating and modifying Ceph Object Gateway zones as archive zones by incorporating the archive tier type into the existing multisite configuration workflow. The Dashboard experience remains consistent with the standard multisite UI, requiring minimal changes to existing workflows. In addition, the Dashboard surfaces archive zone information and operational details, enabling monitoring and management similar to standard zones. As a result, users can configure and manage archive zones directly from the Dashboard, improving usability, consistency, and operational efficiency.

(ISCE-1970)

## Ceph File System (CephFS)

**Added new MDS metrics for improved observability**

Previously, key metrics required for monitoring MDS behavior and performance were not available.

With this enhancement, additional MDS metrics are introduced, improving visibility into metadata operations and helping administrators proactively manage workloads.

(ISCE-3059)

**Optimized CephFS default configurations and recommendations**

Previously, default configurations for MDS and CephFS required manual tuning for optimal performance in different environments.

With this enhancement, improved defaults and configuration recommendations are introduced, helping administrators achieve better performance and stability with minimal tuning.

(ISCE-2701)

**Improved CephFS mirroring performance and reliability**

Previously, CephFS mirroring required performance tuning and lacked optimizations for large-scale replication workloads.

With this enhancement, improvements to mirroring functionality enhance performance evaluation and overall reliability of replication across clusters.

(ISCE-2699)

**Enhanced CephFS subvolume metrics visibility**

Previously, administrators had limited insight into subvolume-level usage and performance characteristics.

With this enhancement, detailed subvolume metrics are introduced for CephFS, enabling improved monitoring, capacity planning, and performance analysis.

(ISCE-3110)

## Ceph Block Device

**Images can now be added to mirrored Ceph Block Device groups without disabling mirroring**

Previously, adding an image to an Ceph Block Device group with mirroring enabled required disabling mirroring, adding the image, and then re-enabling mirroring. As a result, the entire group was resynchronized to the secondary cluster, causing significant network overhead and increased synchronization time.

With this enhancement, images can be added to a mirrored Ceph Block Device group while mirroring remains enabled by using the `rbd group image add` command. Only the newly added image is synchronized to the secondary cluster, while existing images are not resynchronized.

As a result, this reduces unnecessary data transfer, lowers synchronization overhead, and improves operational efficiency.

```
rbd group image add POOL/GROUP POOL/IMAGE
```

(IBMCEPH-13503)

## Ceph Object Gateway

**Enhanced Lua performance using bytecode caching**

Previously, Lua scripts were compiled on every execution, leading to unnecessary overhead and reduced performance.

With this enhancement, Lua bytecode is cached in memory, eliminating repeated compilation and improving execution efficiency for both small and large object workloads.

(IBMCEPH-12115)

**Enhanced Ceph Object Gateway policy evaluation logging and testing**

Previously, visibility into policy evaluation behavior was limited, making it difficult to trace decisions and troubleshoot access issues.

With this enhancement, setting the **`rgw_copious_policy_logging`** parameter to true enables detailed tracing of policy evaluation in logs. In addition, the `rgw-policy-test` command provides verbose policy evaluation from the command line, helping administrators analyze and validate policy behavior more effectively.

(IBMCEPH-8459)

**Enhanced lifecycle processing efficiency for multi-rule configurations**

Previously, lifecycle operations involving multiple overlapping rules required multiple passes over objects, impacting performance in large-scale deployments.

With this enhancement, rules are grouped and evaluated in a single pass, improving performance and scalability for multi-site environments.

(IBMCEPH-12820)

**Enhanced support for `RestrictPublicBuckets` in PublicAccessBlock configuration**

Previously, setting the `RestrictPublicBuckets` option in the PublicAccessBlock configuration had no effect on bucket access behavior because the setting was not implemented.

With this enhancement, support for `RestrictPublicBuckets` is implemented, ensuring that the setting is enforced as expected and improving control over public access restrictions for buckets.

(IBMCEPH-10584)

**Enhanced support for Lua-based abort requests**

Previously, there was no way to abort a Ceph Object Gateway (`rgw`) request based on the result of a Lua script. If a Lua script encountered a syntax or runtime error, Ceph Object Gateway logged the error but continued processing the request.

With this enhancement, Lua scripts can now return the RGW_ABORT_REQUEST error code to explicitly signal that the request should be aborted. Ceph Object Gateway interprets this return value as `-EPERM` and stops processing the request. This return value is evaluated only in the prerequest context and is ignored in other RGW request-processing contexts. As a result, users can implement more effective request control using Lua scripts, enabling conditional request rejection and improved policy enforcement.

(IBMCEPH-13203)

## RADOS

**Added `ok-to-upgrade` command for safer OSD upgrades**

Previously, upgrade orchestration relied on less efficient mechanisms that prolonged upgrade times.

With this feature, the **`ceph osd ok-to-upgrade`** command identifies safe sets of OSDs that can be upgraded simultaneously without risking data availability, improving upgrade efficiency and operational safety.

(IBMCEPH-12147)

**Optimized EC stripe unit alignment for performance**

Previously, stripe unit values could be configured without alignment constraints, leading to suboptimal erasure coding performance.

With this enhancement, stripe unit values must be 4K aligned, ensuring improved performance and efficiency for EC workloads.

(IBMCEPH-13263)

**Added RocksDB performance counters**

Enhanced visibility into BlueStore RocksDB cache behavior Previously, administrators had limited visibility into RocksDB cache usage and shard-level behavior in BlueStore, making it difficult to analyze cache efficiency and diagnose performance issues.

With this enhancement, Red Hat Ceph Storage provides additional performance counters and administrative commands that enable detailed monitoring of RocksDB cache usage, including shard-level statistics. This

improvement helps administrators identify cache inefficiencies and perform more effective performance analysis and tuning.

(ISCE-2956, IBMCEPH-15595)

# Deprecated functionality

This section provides an overview of functionality that has been deprecated in all minor releases up to this release of Red Hat Ceph Storage.

> **Important:** Deprecated functionality continues to be supported until the end of life of Red Hat Ceph Storage 9.x. Deprecated functionality will likely not be supported in future major releases of this product and is not recommended for new deployments. For the most recent list of deprecated functionality within a particular major release, refer to the latest version of release documentation.

**Deprecated method of configuring OIDC federation and IAM roles at the tenant level**

All OIDC resources are now managed as resources within a Ceph Object Gateway account. These OIDC resources include providers, roles, and polices, As a result, all OIDC operations that target a tenant, including the global or empty tenant, are considered deprecated. The deprecated operations incldue creating providers, creating roles, and assuming roles.

With the newer per-account model, federated users are directly associated with the account and Ceph Object Gateway no longer creates *shadow users* (for example, *`TENANT$USER_NAMESPACE`*) upon role assumption. The account itself tracks all resources and identities.

Tenant-based OIDC federation users should migrate their configurations to the new Ceph Object Gateway per-account model, before feature removal.

For more information, see **Red Hat Ceph Storage Ceph Object Gateway › Authenticating with Ceph Object Gateway › Secure Token Service**.

(ISCE-3162)

# Technology Preview

This section provides an overview of Technology Preview features introduced or updated in this release of Red Hat Ceph Storage.

> **Important:** Technology Preview features are not supported with production service level agreements (SLAs), might not be functionally complete, and does not recommend using them for production. These features provide early access to upcoming product features, enabling customers to test functionality and provide feedback during the development process.

## Ceph File System (CephFS)

**Support for CephFS snapshot mirroring**

With this feature you can replicate a CephFS to a remote CephFS on another Ceph storage cluster. Snapshot synchronization copies snapshot data to a remote Ceph File System, and creates a new snapshot on the remote target with the same name. You can configure specific directories for snapshot synchronization.

> **Note:** To use CephFS snapshot mirroring, both the source and the target storage clusters must be running the same Red Hat Ceph Storage version.

For more information, see **Red Hat Ceph Storage File Systems › CephFS snapshot mirroring (Technology Preview)**.

### Ceph Object Gateway

**New per-user and per-bucket usage counters in Prometheus**

Ceph Object Gateway now exports per-user and per-bucket usage counters via performance counters automatically collected by the ceph-exporter and made available in Prometheus. This provides low-overhead, real-time visibility into the following:

- Per-bucket metrics: used bytes, utilized bytes, and number of objects
- Per-user metrics: used bytes and number of objects
- Cache performance metrics: cache hits, misses, updates, and evictions

> **Note:** These metrics are disabled by default. To enable them, configure the appropriate settings in your Ceph Object Gateway configuration.

For more information, see **Red Hat Ceph Storage Dashboard › Managing Ceph Object Gateway › Viewing Ceph Object Gateway per-user and per-bucket performance counters**.

(BZ#2036531)

### RADOS

**Balanced primary placement groups can now be observed in a cluster**

Previously, users could only balance primaries with the offline `osdmaptool`.
With this enhancement, autobalancing is available with the `upmap` balancer. Users can now choose between either the `upmap-read` or `read` mode. The `upmap-read` mode offers simultaneous `upmap` and `read` optimization. The `read` mode can only be used to optimize reads.

For more information, see **Administering Red Hat Ceph Storage › Operations › Managing manager daemons › Using the Ceph Manager balancer module**.

(BZ#1870804)

**Now supports tracking data availability score of a cluster**

This release introduces a feature that tracks the data availability score of a Ceph cluster over time. The score represents how accessible your data is at any given moment, based on factors such as OSD health, placement group states, and redundancy policies.

By monitoring this metric, administrators gain a fact-based view of cluster reliability and can validate availability percentages (for example, 99.99%) against service-level objectives. This capability provides actionable insight into operational resilience and helps ensure confidence in Ceph as a storage platform for critical workloads.

For more information, see **Administering Red Hat Ceph Storage › Monitor Red Hat Ceph Storage clusters › Monitor a Ceph cluster with the CLI › High-level monitoring › Track the data availability score of a cluster (Technology Preview)**.

(ISCE-2144)

## Bug fixes

This section describes bugs with significant user impact, which were fixed in this release of Red Hat Ceph Storage. In addition, the section includes descriptions of fixed known issues found in previous versions.

### `cephadm` utility

**Improved error reporting for OSD redeployment when DB device slots are unavailable**

Previously, the `ceph-volume batch` command did not correctly count devices with existing LVM data. After zapping an OSD, redeployment could fail silently because no DB device slots were available.

With this fix, the condition is treated as an error and a clear message is logged. As a result, when deployment fails due to unavailable DB device slots, users are informed through explicit log errors.

(IBMCEPH-12884)

**Invalid HA cluster port values are now prevented**

Previously, when creating a high availability (HA) cluster with a custom an internal offset caused the assigned port to exceed the valid range.

With this fix, validation has been added to ensure that the resulting port value does not exceed the maximum allowed limit of 65535. As a result, if the provided custom port plus the offset exceeds the valid range, the HA cluster creation command fails with a validation error instead of assigning an invalid port.

(IBMCEPH-11819)

## Ceph Manager

**Manager modules now load reliably after failover**

Previously, a race condition in the manager (MGR) component could cause modules to fail to load in time after a failover. This could happen during an upgrade or when running the **`ceph mgr fail`** command. As a result, MGR module commands could fail with misleading error messages indicating that a module was not enabled or loaded, even though the issue was related to delayed loading.

With this fix, the MGR declares itself active only after all modules have been successfully loaded and are ready to receive commands. In addition, error messages have been improved to clearly distinguish between modules that failed to load in time and those that are not enabled. As a result, MGR module commands now execute reliably after failover, and clearer health warnings are provided if any modules fail to load in time.

(IBMCEPH-9881)

## Ceph Dashboard

**Improved clarity of performance card empty states**

Previously, a generic empty state message was displayed when storage or Prometheus was not configured, providing no clear indication of the issue.

With this fix, empty state messages are displayed dynamically based on the configuration status. As a result, users can clearly identify whether Prometheus is disabled, not configured, storage is not configured, or performance charts are displayed as expected.

(IBMCEPH-13420)

**Duplicate storage class name validation in Ceph Dashboard**

Previously, the Ceph Dashboard did not validate duplicate storage class names during creation. As a result, creating a storage class with a name that already existed unintentionally modified the existing storage class instead of creating a new one.

With this fix, validation is added to prevent duplicate storage class names. As a result, users can no longer create a storage class with a name that already exists, preventing unintended modifications to existing storage classes.

(IBMCEPH-13044)

## Ceph File System (CephFS)

**Quota display is now accurate in mixed quota modes**

Previously, if a quota was set on a parent directory, it was used for displaying quota information. This caused incorrect values when different quota types were defined at different directory levels.

With this fix, the system identifies the correct quota root separately for `max_files` and `max_bytes`. As a result, quotas are displayed correctly in mixed quota configurations.

(IBMCEPH-12286)

**Fixed stability issues in `LogSegment` lifecycle management**

Previously, `LogSegment` objects were managed using raw pointers, allowing multiple MDS components to reference them without a clear ownership model. This could leave dangling pointers when segments were trimmed or expired.

With this fix, reference counting has been introduced for `LogSegment`. As a result, segments are released only after all references are cleared, eliminating dangling pointers and improving MDS stability during operations such as replay and recovery.

(IBMCEPH-11811)

**Snapshot info now correctly includes enctag**

Previously, the `enctag` field was not populated when querying filesystem snapshot information. As a result, an error occurred and the snapshot info output was not fully displayed.

With this fix, the `enctag` field is populated during snapshot queries and snapshot information now displays correctly, including all attributes such as `enctag`.

(IBMCEPH-13575)

**MDS no longer crashes due to improper `mds_lock` usage**

Previously, a bug in the MDS allowed certain code paths to execute without holding the required `mds_lock`, leading to unsafe behavior and unpredictable crashes.

With this fix, the `mds_lock` is consistently held wherever required. As a result, the MDS operates reliably without unexpected crashes.

(IBMCEPH-13700, IBMCEPH-13705)

## Ceph Block Device

**Clone image sparseness is preserved during live migration**

Previously, due to an implementation defect, sparseness was lost when live migrating a clone image. As a result, the destination image could consume significantly more space than the source clone image, including its entire parent chain.

With this fix, the implementation has been corrected to preserve sparseness during live migration. As a result, the destination image now consumes the same amount of space as the source clone image, including its parent chain.

(IBMCEPH-13504)

## Ceph Object Gateway

**Replayed multipart upload requests now return correct `etag`**

Previously, the `CompleteMultipartUpload::etag` member was not assigned when replaying an already completed multipart upload request. As a result, an empty `etag` header was returned for replayed requests.

With this fix, the `etag` member is assigned the validated value in the replay response path. As a result, replayed requests now return the correct `etag` as expected.

(IBMCEPH-13821)

**Conditional multipart upload operations now function as expected**

Previously, while S3 operations were supported, HTTP preconditions for conditional multipart upload were not.

With this fix, HTTP preconditions for conditional multipart upload are fully implemented. As a result, conditional multipart upload operations are now supported as expected.

(IBMCEPH-13557)

**Unordered bucket listings no longer repeat entries or loop indefinitely**

Previously, unordered bucket listing had a bug when traversing regions of the entry space, which could cause entries to repeat or the operation to loop indefinitely.

With this fix, the traversal logic has been corrected. As a result, unordered bucket listing completes as expected without duplicate entries or infinite loops.

(IBMCEPH-12291)

**Ceph Object Gateway no longer crashes during lifecycle processing of versioned buckets**

Previously, the `rgw` daemon could crash during lifecycle (LC) processing on versioned buckets with a large number of object versions, particularly when expiration and noncurrent version expiration rules were applied concurrently.

With this fix, lifecycle processing handles object listing boundaries correctly, preventing invalid memory access during concurrent or batched operations. As a result, Ceph Object Gateway (`rgw`) no longer crashes during lifecycle deletion, and lifecycle processing completes reliably on large versioned buckets.

(IBMCEPH-14308)

**Log buckets are now supported in erasure-coded pools**

Previously, log buckets could not be created in erasure-coded (EC) pools due to append operation limitations.

With this fix, log records are first written to a temporary object in the replicated `default.rgw.log` pool and then copied to the EC pool upon commit. Implicit commit operations are handled asynchronously by a new `BucketLoggingManager`, while explicit commits continue to run synchronously. As a result, log buckets can now be used with EC pools, enabling more flexible storage configurations.

(IBMCEPH-11747)

**Restore operations now validate storage class and return correct results**

Previously, restore operations targeting a non-existent storage class succeeded silently and defaulted to the `STANDARD` class. However, the head-object API still reported the non-existent storage class, creating a mismatch between actual data placement and reported metadata.

With this fix, the restore operation validates whether the requested storage class exists before proceeding and fails with an appropriate error if it does not. As a result, restore operations no longer succeed with invalid storage classes, and object metadata accurately reflects the actual storage class.

(IBMCEPH-11517)

**Cloud restore and transition operations are more robust and reliable**

Previously, multiple issues affected cloud restore and transition workflows. Setting `Days=0` in restore requests could leave objects in a broken state, multipart upload resumption could fail, restored objects could include incorrect ETag formatting, and the restore process was not shut down in the correct order.

With this fix, these issues are addressed across the restore and transition modules to ensure correct request handling, proper ETag formatting, and orderly process management. As a result, cloud restore and transition operations are more reliable, improving overall stability and service availability.

(IBMCEPH-9849)

## Ceph Object Gateway multi-site

**Bucket index metadata remains consistent after lifecycle expiration**

Previously, in rare cases, lifecycle expiration in versioned buckets could leave behind stale bucket index metadata. This occurred due to incorrect shutdown and destruction ordering, which could trigger a segmentation violation and prevent proper cleanup. As a result, stale index entries remained even after objects were removed, potentially impacting operations that rely on accurate bucket index listings and causing errors such as `(27) File too large`.

With this fix, the Asio run queue is drained before destroying the storage backend, ensuring proper shutdown sequencing and cleanup. As a result, bucket index metadata is now correctly maintained after lifecycle expiration, preventing stale entries and improving stability.

(IBMCEPH-12980)

**Ceph Object Gateway no longer crashes on client disconnect and bulk delete operations**

Previously, two issues could cause Ceph Object Gateway to crash during request handling. When a client disconnected while Ceph Object Gateway was sending an error response, an unhandled exception could trigger an abort. In addition, bulk delete requests could cause an assertion failure due to attempts to close an already closed XML section.

With this fix, broken pipe errors from client disconnects are handled gracefully at write time, and redundant XML section close calls are removed. As a result, Ceph Object Gateway no longer crashes during client disconnects or bulk delete operations, improving overall stability.

(IBMCEPH-14447)

## RADOS

**Idle PGs are no longer incorrectly flagged as `stuck peering`**

Previously, when OSDs restarted or were marked down, the cluster could report placement groups (PGs) as `stuck peering` even when they were active. This led to misleading `HEALTH_WARN` messages.

With this fix, PG statistics are updated more frequently, including for idle PGs, and the `last_active` and `last_peered` timestamps and reporting logic are adjusted. As a result, idle PGs are no longer incorrectly

flagged as stuck peering, and `HEALTH_WARN` messages are triggered only after genuine timeouts (greater than 60 seconds).

(IBMCEPH-6744)

## All bug fixes

This section lists a complete listing of all bug fixes in this release of Red Hat Ceph Storage 9.1.

| Issue key | Severity | Summary |
|---|---|---|
| IBMCEPH-12414 | Critical | OSD journal replay failures during startup |
| IBMCEPH-13341 | Critical | Get request fails with 404 after successful HEAD operation |
| IBMCEPH-13575 | Critical | Subvolume info CLI command fails due to undefined variable |
| IBMCEPH-13576 | Critical | Cephadm bootstrap fails with SSH user and custom keys |
| IBMCEPH-13592 | Critical | RGW dashboard delete confirmation text not clearly visible |
| IBMCEPH-13611 | Critical | Dashboard missing archive option during multisite import |
| IBMCEPH-13617 | Critical | Dashboard missing option to edit multisite sync configuration |
| IBMCEPH-13672 | Critical | Object browser page remains loading indefinitely |
| IBMCEPH-13680 | Critical | Bucket creation page stuck without response in dashboard |
| IBMCEPH-13718 | Critical | Clean installation of object browser fails |
| IBMCEPH-13749 | Critical | Per user and per bucket counters not available |
| IBMCEPH-13782 | Critical | Uploaded objects show incorrect file size in dashboard |
| IBMCEPH-13790 | Critical | Dashboard user export creation fails with internal server error |
| IBMCEPH-13799 | Critical | S3 browser not showing buckets with SSL endpoint |
| IBMCEPH-13861 | Critical | Dashboard should prevent login with invalid credentials and remain on login page |
| IBMCEPH-13998 | Critical | Cluster deployment fails when using sudo user |
| IBMCEPH-14005 | Critical | Dashboard sync from option retains old zone after configuration change |
| IBMCEPH-14031 | Critical | Incorrect archive option behavior during multisite setup |
| IBMCEPH-14082 | Critical | Dashboard support diagnostics section shows incorrect entries |
| IBMCEPH-14088 | Critical | Dashboard documentation link returns 404 page |

| Issue key | Severity | Summary |
|---|---|---|
| IBMCEPH-14142 | Critical | Uploaded files not shown immediately in dashboard object view |
| IBMCEPH-14191 | Critical | Dashboard bucket creation stuck after service creation |
| IBMCEPH-14193 | Critical | EC optimization default incorrectly set to false |
| IBMCEPH-14223 | Critical | ceph orch accept call home enabled command not working correctly |
| IBMCEPH-14228 | Critical | Dashboard API missing functionality to set and clear MOTD |
| IBMCEPH-14249 | Critical | Dashboard multisite sync not working as expected |
| IBMCEPH-14255 | Critical | Orchestrator cannot redeploy DB devices |
| IBMCEPH-14267 | Critical | Keepalived service default configuration causes service issues |
| IBMCEPH-14273 | Critical | RGW segfault during AWS cloud transition with HTTPS endpoint |
| IBMCEPH-14278 | Critical | RGW multisite sync daemon crashes during large bucket full sync |
| IBMCEPH-14308 | Critical | RGW crash during lifecycle deletion on versioned buckets |
| IBMCEPH-14314 | Critical | Ceph orch upgrade violates enforced upgrade order |
| IBMCEPH-14377 | Critical | s3 browser refresh shows duplicate entries |
| IBMCEPH-14378 | Critical | RGW not able to configure TLS 1.3 cipher suites using ssl_ciphers parameter |
| IBMCEPH-14441 | Critical | Cephadm bootstrap fails on RHCS 9.1 while configuring dashboard SSL port |
| IBMCEPH-15007 | Critical | cephadm bootstrap fails with crimson image if automatically accept license flag not provided |
| IBMCEPH-15196 | Critical | RGW Cloud Tier Restore fails for AWS S3 with SSE-S3 encryption error |
| IBMCEPH-15708 | Critical | OSDs going down after monitor remove add operations |
| IBMCEPH-12080 | Important | Bootstrap fails due to incorrect IP address validation |
| IBMCEPH-12115 | Important | Performance degradation observed for certain object sizes |
| IBMCEPH-12147 | Important | ok-to-upgrade command behavior needs improvement |

| Issue key | Severity | Summary |
|---|---|---|
| IBMCEPH-12286 | Important | Quota reset not reflected until quota files refreshed |
| IBMCEPH-12701 | Important | RGW segfault occurs during object upload |
| IBMCEPH-12719 | Important | OSD service count misreported in orchestrator output |
| IBMCEPH-13044 | Important | Duplicate storage class creation overwrites existing configuration |
| IBMCEPH-13315 | Important | Cephadm SSH channel hardening incomplete |
| IBMCEPH-13318 | Important | Directory configuration issue impacts namespace behavior |
| IBMCEPH-13340 | Important | Required backport request needed for upstream fix |
| IBMCEPH-13401 | Important | Firewall port not opened for HA proxy peer communication |
| IBMCEPH-13403 | Important | Required backport missing in release build |
| IBMCEPH-13450 | Important | Optional service dependency installation missing during host preparation |
| IBMCEPH-13491 | Important | RGW S3 requests return HTTP 409 error |
| IBMCEPH-13494 | Important | HeadObject request returns incorrect 403 response |
| IBMCEPH-13540 | Important | Bootstrap CLI help message duplication for license options |
| IBMCEPH-13557 | Important | Multipart upload handling requires fix |
| IBMCEPH-13594 | Important | Unable to modify multisite endpoints in dashboard |
| IBMCEPH-13605 | Important | Dashboard landing page contains excessive empty space |
| IBMCEPH-13621 | Important | Duplicate capacity metrics displayed for CephFS in dashboard |
| IBMCEPH-13639 | Important | CephFS mirror directories stuck in syncing state |
| IBMCEPH-13651 | Important | Dashboard Grafana capacity metrics not displayed correctly |
| IBMCEPH-13662 | Important | RGW crashes when admin sends invalid AssumeRole request |
| IBMCEPH-13669 | Important | Object browser login page missing example inputs and guidance |
| IBMCEPH-13784 | Important | Creating folder generates unintended dummy file |
| IBMCEPH-13788 | Important | RGW multisite full sync fails silently while reporting success |
| IBMCEPH-13794 | Important | Dashboard redirects to add storage page after re login |

| Issue key | Severity | Summary |
|---|---|---|
| IBMCEPH-13851 | Important | RGW segfault occurs under high S3 write workload |
| IBMCEPH-13864 | Important | Edit client config option stuck in loading state |
| IBMCEPH-13894 | Important | Dashboard device availability warnings are misleading |
| IBMCEPH-13921 | Important | Read only users cannot access dashboard overview page |
| IBMCEPH-13931 | Important | Dashboard filesystem overview missing MDS details |
| IBMCEPH-13973 | Important | Dashboard shows RAM disks incorrectly in device inventory |
| IBMCEPH-14053 | Important | Bucket creation remains stuck when bucket already exists |
| IBMCEPH-14054 | Important | Object browser image should use ICR image instead of current image |
| IBMCEPH-14059 | Important | Bucket settings show incorrect object count instead of actual objects |
| IBMCEPH-14135 | Important | Dashboard missing MDS observability metrics |
| IBMCEPH-14140 | Important | Dashboard reports incorrect CephFS capacity usage |
| IBMCEPH-14207 | Important | Dashboard pool manager role unable to create pools due to access denied |
| IBMCEPH-14282 | Important | Dashboard icon misalignment in add storage and overview buttons |
| IBMCEPH-14429 | Important | Unable to recreate topic created with invalid characters |
| IBMCEPH-11602 | Moderate | Directory removal implementation inefficient |
| IBMCEPH-11846 | Moderate | Dashboard table layout expands incorrectly when selecting modules |
| IBMCEPH-11914 | Moderate | Dashboard replication policy changes not applied correctly |
| IBMCEPH-12403 | Moderate | Deleting storage class does not remove associated zone entry |
| IBMCEPH-12520 | Moderate | Dashboard metrics not visible after upgrade |
| IBMCEPH-12659 | Moderate | Deep-scrub completion timing inconsistent |
| IBMCEPH-12689 | Moderate | OSD deployment fails on nodes with high OSD count |
| IBMCEPH-12820 | Moderate | Replication for new buckets does not function correctly |
| IBMCEPH-12873 | Moderate | Linking bucket to account fails with internal server error |

| Issue key | Severity | Summary |
|---|---|---|
| IBMCEPH-12987 | Moderate | Dashboard unable to restart or stop RGW service |
| IBMCEPH-13263 | Moderate | Stripe unit alignment restriction missing for EC optimization |
| IBMCEPH-13555 | Moderate | OSD delete popup missing option to preserve OSD ID |
| IBMCEPH-13750 | Moderate | Dashboard empty state illustration missing image asset |
| IBMCEPH-13890 | Moderate | Dashboard missing support for `cephadm` managed SSL certificates |
| IBMCEPH-14089 | Moderate | Dashboard overview graphs display incorrect decimal values |
| IBMCEPH-14144 | Moderate | Dashboard bucket content not visible without manual refresh |
| IBMCEPH-13731 | Low | Dashboard host selection not populated during namespace creation |
| IBMCEPH-13785 | Low | Dashboard shows duplicate objects intermittently in object browser |

# Known issues

This section documents known issues found in this release of Red Hat Ceph Storage.

### `cephadm` utility

**Promtail image continues to appear in `cephadm` list-images output**

The Promtail container image might still appear in `cephadm` list-images output after upgrading to Alloy.

Currently, during the transition from Promtail to Alloy, `cephadm` continues to register the Promtail container image to maintain backward compatibility. As a result, the Promtail image remains visible even though Alloy is the default logging solution.

As a workaround, no action is required. Ignore the Promtail image entry during the transition phase.

(IBMCEPH-13162)

**Grafana certificate does not migrate during upgrade**

When you upgrade from Red Hat Ceph Storage 8.1 to 9.0, the existing user-signed Grafana certificate is not migrated. Instead, Grafana switches to a `cephadm`-signed certificate. As a result, duplicate certificate entries may appear, and certificate-related health warnings can persist. Manual reconfiguration is required if you want to use custom TLS certificates.

> **Note:** Data services remain unaffected.

To work without custom TLS certificates, you can continue using the `cephadm`-signed certificate.

As a workaround to use custom TLS certificates, complete the following steps:

1. Change the Grafana specification to use `certificate_source: reference`.
2. Use `certmgr` to upload a valid user-signed certificate and key for each host.
3. Run the **`ceph orch reconfig grafana`** command.

(IBMCEPH-13080)

**Monitor configuration updates are not applied on restart**

When you update monitor configuration settings, such as `public_network`, the changes are not applied when the monitor daemon is restarted. This occurs because monitor daemon configurations are not dynamically refreshed during a restart.

As a result, the monitor continues to run with the previous configuration, and the updated values do not take effect.

As a workaround, redeploy the monitor daemon instead of restarting it after updating the configuration. This ensures that the updated configuration is applied successfully.

(IBMCEPH-12242)

**Crash daemon cannot access crash directory due to permission changes**

When certain services, such as Grafana, are deployed, the permissions of the crash directory can change. As a result, the crash daemon cannot access the directory, preventing it from functioning correctly.

As a workaround, manually update the permissions of the crash directory to `167`. You must repeat this action each time a daemon deployment changes the directory permissions to ensure proper access.

(IBMCEPH-12678)

## Ceph File System (CephFS)

**`root_squash` kernel client may cause data inconsistency and triggers `HEALTH_ERR`**

A bug in the root_squash implementation can cause changes made by a kernel client restricted with `root_squash` capabilities to be lost. Although the issue is fixed for the FUSE client and the MDS, the kernel client remains affected. As a result, the cluster emits the following error when it detects a client with the broken `root_squash` implementation:

```
HEALTH_ERR: MDS_CLIENTS_BROKEN_ROOTSQUASH
```

This occurs, due to the risk of data inconsistency and lost updates.

To avoid this issue, it is recommended to discontinue using `root_squash` with kernel clients until a fix is available.

To prevent affected clients from connecting, you can evict and permanently block them by setting the required client feature.

```
ceph fs required_client_features add client_mds_auth_caps
```

This helps protect the cluster from inconsistent behavior caused by affected clients.

(IBMCEPH-14902)

## Ceph Object Gateway multi-site

**Secondary site displays old zonegroup name after rename**

After renaming a zonegroup in a multisite configuration, the secondary site might still display the previous zonegroup name.

Currently, when a zonegroup is renamed on the primary site, the old name is not removed from the .rgw.root pool. As a result, both the old and new zonegroup names appear in the `radosgw-admin zonegroup list` output, and sync operations might be impacted.

As a workaround, perform the following steps:

1. Verify that the new zonegroup name exists.

   ```
   radosgw-admin zonegroup list
   ```

2. List entries in the `.rgw.root` pool and locate the old zonegroup name.

   ```
   rados -p .rgw.root ls
   ```

   The old name appears in the format:

   ```
   zonegroups_names.OLD_ZONEGROUP_NAME
   ```

3. Remove the old zonegroup name from the pool:

```
rados -p .rgw.root rm zonegroups_names.OLD_ZONEGROUP_NAME
```

Removing the old zonegroup name restores normal sync operations.

(IBMCEPH-13140)

### RADOS

**`ceph versions -f xml` command produces non-well-formed XML output**

When you run the **`ceph versions -f xml`** command, the generated output is not well-formed XML and cannot be parsed by standard XML parsers. This occurs because the command uses full Ceph version strings (including special characters such as dots, parentheses, spaces, and hashes) as XML tag names, which violates XML syntax rules.

As a result, XML parsing fails with errors such as not well-formed (invalid token), preventing automated processing or validation of the output.

Currently there is no workaround.

(IBMCEPH-13690)

## Sources

Use this information to attain Red Hat Ceph Storage source code packages.

The updated Red Hat Ceph Storage source code packages are available at the following locations:

- For Red Hat Enterprise Linux 9: http://ftp.redhat.com/redhat/linux/enterprise/9Base/en/RHCEPH/SRPMS/
- For Red Hat Enterprise Linux 10: https://ftp.redhat.com/redhat/linux/enterprise/10Base/en/RHCEPH/SRPMS/

# Asynchronous updates

This section describes the bug fixes, known issues, and enhancements for asynchronous updates.

## Release notes for 9.1z1

An update is now available for Red Hat Ceph Storage 9.1z1 which provides bug fixes and a new known issue.

### New features and enhancements

This section lists all the major updates, and enhancements introduced in this release of Red Hat Ceph Storage.

#### Security

**New support for CephX key type**

Ceph now supports CephX aes256k keys. `cephadm` automatically rotates the keys of the service daemons with CephX aes256k key type.

For more information, see **Red Hat Ceph Storage Overview › Data security and hardening › Encryption and key management › Enabling key rotation › CephX key rotation**.

### Bug fixes

This section describes bugs with significant user impact, which were fixed in this release of Red Hat Ceph Storage. In addition, the section includes descriptions of fixed known issues found in previous versions.

#### Ceph Dashboard

**Dashboard no longer displays a 500 error when retrieving upgrade image information**

Previously, the Dashboard Upgrade API could fail to retrieve image information. As a result, users encountered a `500 - Internal Server Error` with the message, `The server encountered an unexpected condition which prevented it from fulfilling the request`, when logging in to the dashboard and accessing the **Overview** page.

With this fix, the Upgrade API retrieves upgrade image information correctly and no longer fails during dashboard operations. As a result, users can access the **Overview** page without encountering the 500 error.

(IBMCEPH-14087)

### All bug fixes

This section lists a complete listing of all bug fixes in this release of Red Hat Ceph Storage 9.1z1.

*Table 2: List of all bug fixes*

| Issue key | Severity | Summary |
|---|---|---|
| IBMCEPH-14087 | Critical | RH Ceph cluster throws 500 Internal Server Error when accessing overview page |
| IBMCEPH-16358 | Critical | Unable to install ceph-common package due to missing rpms |
| IBMCEPH-16369 | Critical | cephfs going to laggy state after upgrade |
| IBMCEPH-16404 | Critical | Disable ceph orch rotate-key command |
| IBMCEPH-16407 | Critical | Do not wipe rotating keyrings during upgrade |

| Issue key | Severity | Summary |
|---|---|---|
| IBMCEPH-16408 | Critical | Ceph Cluster Upgrade Failure. Modules Later Alphabetically Than "mirroring" Fail to Initialize After Mgr Upgrades |
| IBMCEPH-16473 | Critical | cephadm OSD daemon fails with handle_auth_bad_method Permission denied |
| IBMCEPH-16475 | Critical | Allow any label in encryption keys start and end markers |
| IBMCEPH-16491 | Critical | Cluster upgrade 8.1 → 9.1z1 fails with UPGRADE_EXCEPTION: local variable 'daemon_spec' referenced before assignment |
| IBMCEPH-16496 | Critical | MGR addition post upgrade fails - CEPHADM_DAEMON_PREPARE_CREATE_FAIL - Unable to fetch keyring |
| IBMCEPH-16497 | Critical | Ceph auth get-or-create command fails for Client keyring |
| IBMCEPH-16513 | Critical | Grafana upgrade is stuck during staggered upgrade of Mgr daemon |
| IBMCEPH-16581 | Critical | Cluster Upgrade fails for OSDs backed by raw devices |
| IBMCEPH-16014 | Important | cephadm parallel deploy/remove can result in multiple identical ganeshas running at the same time causing EIO in client apps |
| IBMCEPH-16017 | Important | Ceph 9.1 build with CephX key type support has old auth dump-keys format |
| IBMCEPH-16385 | Important | ceph mon set usage is missing cipher related command descriptions |

## Security fixes

This section lists security fixes from this release of Red Hat Ceph Storage 9.1z1.

There are no CVE security fixes for this release. For details about CVEs, see CVE Records.

## Known issues

This section documents known issues found in this release of Red Hat Ceph Storage.

### `cephadm` utility

#### `cephadm` host checks fail in FIPS mode when using ED25519 SSH keys

When FIPS mode is enabled on cluster nodes, `cephadm` host checks can fail if the stored cephadm SSH identity uses an ED25519 key. Because ED25519 keys are not supported in FIPS mode, `cephadm` cannot import the SSH private key and reports a `KeyImportError('Unknown OpenSSH private key algorithm')` error. As a result, managed hosts are marked `Offline` and the cluster raises a `CEPHADM_HOST_CHECK_FAILED` health warning.

As a workaround, replace the `cephadm` ED25519 SSH identity with a FIPS-compatible RSA key.

Verify whether `cephadm` is using an ED25519 key.

```
ceph cephadm get-pub-key
```

If the output starts with `ssh-ed25519`, continue with the following steps.

1. Generate an RSA key pair on the administration host.

   ```
   ssh-keygen -t rsa -b 4096 -f /root/cephadm_rsa -N ''
   ```

2. Install the RSA public key on all Ceph hosts.

   a. On the administration host, run the following commands:

   ```
   mkdir -p /root/.ssh
   chmod 700 /root/.ssh
   cat /root/cephadm_rsa.pub >> /root/.ssh/authorized_keys
   chmod 600 /root/.ssh/authorized_keys
   ```

   b. Copy the key to each Ceph host.

   ```
   ssh-copy-id -f -i /root/cephadm_rsa.pub root@HOST_IP
   ```

3. Verify that RSA-based SSH access works.

   ```
   ssh -i /root/cephadm_rsa \
   -o IdentitiesOnly=yes \
   root@HOST_IP 'hostname; fips-mode-setup --check; cat /proc/sys/crypto/
   fips_enabled'
   ```

   Expected output should confirm the hostname and FIPS state.

   ```
   FIPS mode is enabled
   ```

4. Update the cephadm SSH identity stored in the configuration key store.

   ```
   cat /root/cephadm_rsa | cephadm shell -- ceph config-key set mgr/cephadm/
   ssh_identity_key -i -
   cat /root/cephadm_rsa.pub | cephadm shell -- ceph config-key set mgr/cephadm/
   ssh_identity_pub -i -
   ```

   Expected output should confirm the hostname and FIPS state.

5. Verify that `cephadm` is using the RSA key.

   ```
   cephadm shell -- ceph config-key get mgr/cephadm/ssh_identity_pub
   ```

   The output should start with `ssh-rsa`.

6. Failover or reload the active manager daemon so that `cephadm` reloads the updated SSH identity.

7. Verify the recovery.

   ```
   ceph orch host ls
   ceph orch ps
   ceph health detail
   ```

After `cephadm` begins using the RSA key, host connectivity is restored and host checks complete successfully.

(IBMCEPH-16272)

## Release notes for 9.1z2

An update is now available for Red Hat Ceph Storage 9.1z2 which provides bug fixes and a new known issue.

# New features and enhancements

This section lists all the major updates, and enhancements introduced in this release of Red Hat Ceph Storage.

### `cephadm` utility

#### `cephadm` now supports configurable persistent logging levels

Previously, `cephadm` used a fixed logging verbosity level for persistent logs written to `/var/log/ceph/cephadm.log` and `syslog`, when enabled.

With this enhancement, you can control the verbosity of `cephadm` persistent logging by using the `mgr/cephadm/CEPHADM_BINARY_LOGGING_LEVEL` configuration option. As a result, administrators can adjust `cephadm` logging verbosity to match operational and troubleshooting requirements, while console output for bootstrap and other interactive operations remains unchanged.

Supported *`CEPHADM_BINARY_LOGGING_LEVEL`* values are `debug`, `info`, `warning`, and `error`. The default value is `debug`.

(IBMCEPH-15698)

### NFS service

#### Enhanced log messages in the NFS share creation path

This release introduces improved log messages in the `create_export` path. Log messages are standardized, previously missing messages are added, and additional context is included to help administrators identify the cause of NFS share creation failures.

(IBMCEPH-17005)

#### New NFS cluster authentication key rotation

Previously, rotating authentication keys for an NFS cluster required manual management of individual keyrings and associated services, increasing administrative effort and the risk of configuration inconsistencies.

With this enhancement, the **`ceph nfs cluster rotate-key`** command enables administrators to rotate authentication keys for an entire NFS cluster or for specific cluster entities. The command supports custom key types through the **`--key-type`** option. It automatically updates associated CephFS share keyrings and redeploys NFS services when daemon keyrings are rotated. This simplifies key management and helps maintain secure NFS deployments.

For more information, see **Red Hat Ceph Storage Overview › Data security and hardening › Encryption and key management › Enabling key rotation › CephX key rotation**.

(IBMCEPH-16471)

# Bug fixes

This section describes bugs with significant user impact, which were fixed in this release of Red Hat Ceph Storage. In addition, the section includes descriptions of fixed known issues found in previous versions.

### Security

#### SigV4 request validation is now enforced for Ceph Object Gateway

Previously, Ceph Object Gateway accepted SigV4 requests even when the `host` and `x-amz-*` headers were not included in the signed header set. This behavior was addressed as part of the fix for CVE-2026-54330. However, the multisite REST client generated requests with these improperly signed headers.

With this fix, Ceph Object Gateway rejects SigV4 requests that do not include the required `host` and `x-amz-*` headers in the signed header set. As a result, SigV4 request validation complies with security requirements.

> **Important:** For multisite deployments, set `rgw_sigv4_insecure` to `true` and use the `--force` option before starting the upgrade. After all clusters have been upgraded, set `rgw_sigv4_insecure` back to `false` on all clusters.

(IBMCEPH-15856)

### `cephadm` utility

#### Monitor configuration updates are now applied when the monitor daemon is restarted

Previously, monitor daemon configurations were not dynamically refreshed during a restart. As a result, updated settings such as **`public_network`** did not take effect, and the monitor continued to run with the previous configuration.

With this fix, monitor configuration updates are correctly applied when the monitor daemon is restarted. As a result, updated configuration values take effect without requiring manual redeployment.

(IBMCEPH-15002)

#### Host management and service placement now work correctly with uppercase hostnames after upgrade

Previously, hostname normalization changes converted hostnames to lowercase during upgrade. In clusters that were originally deployed with uppercase hostnames, existing cache dictionaries remained keyed with the original uppercase values, causing hostname lookups to fail.

With this fix, hostname normalization is handled correctly for clusters that contain uppercase hostnames created before the upgrade. As a result, host management operations and service placement work as expected after upgrade, and errors such as `Failed to apply <n> service(s)` and `host <hostname> does not exist` no longer occur.

(IBMCEPH-16538)

#### `mgmt-gateway` services remain stable when using `cephadm`-signed certificates

Previously, when the `mgmt-gateway` service was configured with `certificate_source: cephadm-signed`, `cephadm` did not include the service specification when calculating persisted dependencies. As a result, the persisted and recalculated dependency lists did not match, causing cephadm to repeatedly reconfigure and redeploy the `mgmt-gateway` service. This behavior could cause the service to become unavailable.

With this fix, `cephadm` correctly includes the service specification when calculating and persisting dependencies for `mgmt-gateway`. The same correction also applies to `oauth2-proxy`. As a result, dependency lists remain consistent when `certificate_source: cephadm-signed` is configured, reducing unnecessary service reconfiguration and helping `mgmt-gateway` remain stable after deployment.

(IBMCEPH-17209)

#### External TLS certificates remain compatible across upgrades

Previously, cephadm could encounter issues when processing externally managed TLS certificate bundles that included intermediate certificates. In some cases, certificate validation could fail, and existing Ceph Object Gateway deployments that used legacy certificate settings could encounter issues after an upgrade.

With this fix, cephadm correctly processes complete TLS certificate bundles and supports migration of valid legacy Ceph Object Gateway certificate configurations. As a result, external TLS certificates are handled more reliably, and existing Ceph Object Gateway deployments remain compatible across upgrades.

(IBMCEPH-16113)

## Ceph Manager

#### Configuration commands now handle whitespace in section names correctly

Previously, `ceph config set`, `ceph config get`, and `ceph config rm` did not trim leading or trailing whitespace from the section name before parsing it. As a result, configuration values set for targets such as `osd.0` or `osd.0 ` could be stored under an unintended key and fail to apply to the expected daemon.

With this fix, leading and trailing whitespace is removed from section names before they are parsed and matched. As a result, configuration commands correctly treat `osd.0`, `osd.0`, `osd.0 `, `osd.0` as the same target, ensuring that configuration values are applied, retrieved, and removed as expected.

(IBMCEPH-10858)

#### OSD redeployment now reuses available DB slots correctly

Previously, when redeploying an OSD with DB devices after a `ceph orch osd rm --zap --replace` operation, `ceph-volume` could incorrectly determine that no DB slots were available on an existing fast device volume group, even when a free slot was available. As a result, the OSD was not provisioned successfully and remained in a destroyed state.

With this fix, `ceph-volume` more accurately identifies available DB slots on existing fast device volume groups during OSD redeployment. Additional improvements were made to device allocation and drive selection logic to address related edge cases. As a result, OSD redeployment correctly reuses available DB slots on existing fast device volume groups, allowing the OSD to be provisioned successfully without manual intervention.

(IBMCEPH-15155)

## Ceph Dashboard

**Dashboard Export Multi-site Realm Token feature now works correctly on secondary sites**

Previously, on secondary sites, attempts to fetch primary zone parameters returned ENOENT, causing the `/api/rgw/realm/get_realm_tokens` API to return HTTP `500`. As a result, the Dashboard Export Multi-site Realm Token feature failed for replicated realms, returning either an empty list or an internal server error.

With this fix, zone parameter retrieval on secondary sites falls back to locally available parameters when the primary zone is unavailable. As a result, the `/api/rgw/realm/get_realm_tokens` API correctly returns realm information and export tokens for replicated realms on both primary and secondary sites.

(IBMCEPH-16585)

**Dashboard processing now handles malformed progress event references safely**

Previously, the Ceph Dashboard could raise an exception when processing progress events because the `refs` field could contain unexpected or malformed data. As a result, the dashboard was unable to process progress events when `refs` was not in the expected format.

With this fix, validation was added for the `refs` field to safely handle dictionaries, valid key-value lists or tuples, `None`, and unexpected values. As a result, the dashboard can process progress events more reliably and avoids exceptions caused by malformed `refs` data.

(IBMCEPH-16427)

## Ceph File System (CephFS)

**Retention field now validates time designators correctly**

Previously, the `Retention` field accepted invalid duration values that did not include a valid time designator format. As a result, values such as 5Y were incorrectly accepted.

With this fix, strict validation is applied to the `Retention` field time designator. As a result, invalid values such as 5Y are rejected, and only correctly formatted retention durations are accepted.

(IBMCEPH-13004)

## Ceph Block Device mirroring

**Ceph Block Device mirror snapshots are no longer pruned after image failover**

Previously, after a Ceph Block Device image failover, obsolete primary and primary-demoted mirror snapshots were not removed until a subsequent promote or demote operation. As a result, unnecessary mirror snapshots accumulated on the secondary cluster and consumed additional storage, with the greatest impact on large images.

With this fix, the `rbd-mirror` daemon proactively removes obsolete primary and primary-demoted mirror snapshots immediately after a successful image failover. As a result, obsolete mirror snapshots are reclaimed sooner, reducing unnecessary storage usage on the secondary cluster.

(IBMCEPH-16990)

**Mirror group snapshot scheduler is now FIPS compliant**

Previously, the mirror group snapshot scheduler used MD5 hashing to calculate deterministic schedule phase offsets. Although the hash was not used for security purposes, MD5 operations were blocked in FIPS mode.

With this fix, the MD5 hash used for phase offset calculation is explicitly marked as not being used for security.As a result, the mirror group snapshot scheduler operates correctly in FIPS-enabled environments and is FIPS compliant.

(IBMCEPH-17724)

## Ceph Object Gateway

### Topic queue reservation leak no longer causes false ENOSPC errors

Previously, the **urgent_data.reserved_size** field accumulated incorrect values over time due to a mismatch between the size added during `reserve()` and the size subtracted during `commit()` and `abort()`. As a result, **urgent_data.reserved_size** grew unbounded, eventually reaching the queue capacity limit and returning ENOSPC errors even when the queue had sufficient available space.

With this fix, reservation calculations are corrected so that the reservation value no longer leaks. In addition, a one-time self-healing mechanism recalculates and corrects the reservation counter on existing topic queues. As a result, topic queues return ENOSPC errors only when they reach their actual capacity limit, and existing queues with accumulated reservation leaks are automatically corrected.

(IBMCEPH-17438)

### Ceph Object Gateway stability is improved during lifecycle and garbage collection processing

Previously, a race condition in asynchronous OMAP read processing could cause Ceph Object Gateway daemons to terminate unexpectedly under heavy workloads. Repeated daemon crashes could interrupt Lifecycle (LC) and Garbage Collection (GC) operations, leading to delayed cleanup and growth of the garbage collection queue.

With this fix, asynchronous worker task handling is coordinated correctly to ensure operations complete safely before resources are released. As a result, Ceph Object Gateway remains stable during concurrent OMAP read operations, allowing LC and GC processing to continue without interruption and preventing garbage collection queue buildup.

(IBMCEPH-17569)

## Multi-site Ceph Object Gateway

### Empty shard trimming no longer loops indefinitely

Previously, a misplaced exception handler inside a loop could cause trimming operations on an empty shard to enter an infinite loop.

With this fix, the exception handling logic is moved outside the loop. As a result, trimming an empty shard completes correctly and no longer loops indefinitely.

(IBMCEPH-15576)

## RADOS

### Crash daemon now retains correct access to the crash directory after service deployments

Previously, deploying certain services such as Grafana changed the permissions of the crash directory path, preventing the crash daemon from accessing the directory and functioning correctly.

With this fix, the crash directory permissions are correctly preserved when services are deployed. As a result, the crash daemon retains access to the crash directory path without requiring manual intervention after each deployment.

(IBMCEPH-14999)

### Ceph Monitors remain stable during multiple monitor removals

Previously, when multiple monitors were removed from the monmap in a single update, the monitor rank adjustment logic could fail to maintain consistency between the local rank and the recalculated rank in the monmap. This issue could trigger an assertion failure and cause a Ceph Monitor to crash.

With this fix, the monitor uses the recalculated rank provided by the monmap instead of attempting to adjust the rank iteratively during monitor removal processing. As a result, Ceph Monitors can process updates that remove multiple monitors in a single monmap update without triggering this assertion failure.

(IBMCEPH-15723)

## NFS service

### CIDR prefix length `/0` is now accepted in NFS share client blocks

Previously, the CIDR suffix lexer rejected a prefix length of `0` because the regex pattern required the first digit to be in the range `1`–`9`. As a result, valid CIDR notations such as `0.0.0.0/0` (RFC 4632) and `::/0` (RFC 4291) could not be used in NFS share client blocks to match all addresses in an address family.

With this fix, the regex accepts a prefix length of `/0` while still disallowing leading zeros in other positions. As a result, `0.0.0.0/0` and `::/0` are now valid in NFS share client block definitions.

(IBMCEPH-15732)

**NFS server metrics are now exposed by default**

Previously, NFS server metrics were not exposed through the HTTP metrics endpoint because metrics collection was not enabled in the `ganesha.conf` configuration.

With this fix, the `Enable_Metrics` setting is enabled in `ganesha.conf` file. As a result, NFS metrics are exposed by default through the HTTP metrics endpoint on port 9587 (`http://LOCAL_IP:9587`).

(IBMCEPH-17615)

**NFS service upgrades now complete without QoS type compatibility errors**

Previously, upgrades could fail because the **`qos_type`** parameter changed from an integer value to a string value in a previous release. As a result, the NFS service daemon could crash when processing the parameter during an upgrade.

With this fix, the NFS service accepts both integer and string values for **`qos_type`**. As a result, NFS service upgrades complete successfully without daemon crashes caused by **`qos_type`** compatibility issues.

(IBMCEPH-16314)

# All bug fixes

This section lists a complete listing of all bug fixes in this release of Red Hat Ceph Storage 9.1z2.

*Table 3: List of all bug fixes*

| Issue key | Summary |
|---|---|
| IBMCEPH-8791 | Unable perform minor upgrade with "ceph orch upgrade start registry.redhat.io/rhceph/rhceph-7-rhel9:latest" |
| IBMCEPH-10858 | cephadm config set allows to set targets with spaces to the right. This leads to confusion and should be trimmed. |
| IBMCEPH-11807 | Upgrade of OSD damons to IBM Ceph 8.1 are failing |
| IBMCEPH-13004 | Add validation for Retention field time designator, retention add 5Y succeeds but not expected |
| IBMCEPH-14271 | Bump Grafana in next release |
| IBMCEPH-14320 | ceph fs perf stats -f json-pretty does not produce expected pretty output |
| IBMCEPH-14898 | Reduce repetitive INFO-level logs from health checks |
| IBMCEPH-14999 | Daemon crashes observed, but crash files are not generated in the expected directory |
| IBMCEPH-15002 | Mon daemons do not get updated with public_network set via config set upon restart |
| IBMCEPH-15155 | OSD redeploy fails when fast device VG is partially used |
| IBMCEPH-15161 | Upstream Reef Documentation shows that we should be able to pause deletes and clone operations but it does not work. |
| IBMCEPH-15576 | Infinite trim on single-site clusters |
| IBMCEPH-15698 | Disabling unwanted logging in the cephadm.log |
| IBMCEPH-15723 | Ceph MON crash in ConnectionTracker::notify_rank_removed(int, int) function |

| Issue key | Summary |
|---|---|
| IBMCEPH-16007 | Ceph MGR crash ** Caught signal (Segmentation fault) ** "in thread 7f6aaa1ab640 thread_name:mgr-fin" |
| IBMCEPH-16020 | OSDMap: FAILED ceph_assert(num_down_in_osds <= num_in_osds) |
| IBMCEPH-16021 | Replication(Time) Delta per shard showing No Data |
| IBMCEPH-16079 | Cephadm not creating OSD daemons for few OSDs despite being present in CRUSH map |
| IBMCEPH-16113 | RGW Daemons removed from cluster while upgrading from 8.1z3 to 9.0z2 |
| IBMCEPH-16131 | MGR prometheus module crashes with CherryPy ChannelFailures AttributeError when ms_inject_socket_failures is set globally |
| IBMCEPH-16295 | inventory fails on hosts with LVM raid mirror volumes |
| IBMCEPH-16402 | startup crash due to race between lifecycle and quota handler |
| IBMCEPH-16427 | Ceph Dashboard menus fail with HTTP 500 error |
| IBMCEPH-16538 | hostname changed to lower case post upgrade to 9.1z2 causing the running service failed to get placed |
| IBMCEPH-16554 | Not able to set motd on FIPS enabled OS clusters |
| IBMCEPH-16585 | Export Multi-site Realm Token in non-master zone does not show token details |
| IBMCEPH-16596 | Fix incorrect rollback logic for partial write OI (9.1zX) |
| IBMCEPH-16990 | rbd mirror not pruning snaps on non primary after fail over |
| IBMCEPH-16991 | Make mirror group promote and demote crash-consistent |
| IBMCEPH-16993 | Remove undo code path in mirror::GroupEnableRequest |
| IBMCEPH-16997 | ceph orch ls crashes when reloading node-proxy service spec |
| IBMCEPH-17009 | RGW with haproxy in concentrator mode fails 'monitor_ips' is undefined |
| IBMCEPH-17023 | Update Required Fields for User Creation in User Management |
| IBMCEPH-17101 | Observing 500 internal error while performing bucket create post delete and recreate User/account metadata cache in mgr after |
| IBMCEPH-17165 | mon crash with SIGABRT in OSDMap::pgtemp_undo_primaryfirst() |
| IBMCEPH-17202 | Zone creation fails while creating from UI |
| IBMCEPH-17205 | Health Warning is missing the descriptive text |
| IBMCEPH-17209 | mgmt-gateway continuously cycling on 9.9.1.0 code |
| IBMCEPH-17218 | RHCS cluster upgrade stalls after first mgr daemon upgrade |
| IBMCEPH-17279 | CephFS unavailable for 5 minutes during MDS upgrade |

| Issue key | Summary |
|---|---|
| IBMCEPH-17287 | OSD crash in ReplicatedBackend::do_repop during rolling upgrade from Squid 19.2.1-375 to Tentacle 20.1.0-221 |
| IBMCEPH-17400 | cephadm: stale config_daemon_deps entries in host cache |
| IBMCEPH-17438 | MOD RGW persistent notification queue reports full when empt bug #74713 |
| IBMCEPH-17463 | Bucket creation via S3 Browser 1.0.0 got stuck with an error "InvalidLocationConstraint: The default" error, even when the correct "shared" region is provided during login |
| IBMCEPH-17474 | used_bytes is not updated after quota reset to 0 and IO |
| IBMCEPH-17541 | s390x PG Inconsistent omap digest mismatch |
| IBMCEPH-17569 | RGW crashes during async shard OMAP reads on versioned buckets |
| IBMCEPH-17571 | upgrade 9.0→9.1 with use_agent=true fails after mgr upgrade |
| IBMCEPH-17724 | RBD Mirror Group Snapshot Scheduler Fails on EC RDR - FIPS-Enabled Clusters Due to MD5 Usage |
| IBMCEPH-17771 | OSD enters error state post Monitor connection scores tests |
| IBMCEPH-17783 | Permission Denied error is encountered while performing a realm pull, blocking the sync between the non-master and master zones |
| IBMCEPH-17813 | SigV4 fix breaks presigned PUT URLs |
| IBMCEPH-17962 | Upgrade Allows Legacy `aes` Cipher Despite `aes256k` Support, Effectively Downgrading Cluster Cipher Policy |
| IBMCEPH-18008 | RGW improperly requires x-amz-sha256 to be signed in S3 requests |

## Security fixes

This section lists security fixes from this release of Red Hat Ceph Storage 9.1z2.

For details about CVEs, see CVE Records.

- CVE-2021-43565
- CVE-2022-27191
- CVE-2023-48795
- CVE-2025-15599
- CVE-2025-26791
- CVE-2025-62718
- CVE-2025-66030
- CVE-2025-66412
- CVE-2026-1933
- CVE-2026-22036
- CVE-2026-27970

- CVE-2026-32141
- CVE-2026-32635
- CVE-2026-33228
- CVE-2026-33891
- CVE-2026-33894
- CVE-2026-33895
- CVE-2026-33896
- CVE-2026-40175
- CVE-2026-40895
- CVE-2026-41238
- CVE-2026-41239
- CVE-2026-41240
- CVE-2026-41305
- CVE-2026-41680
- CVE-2026-42033
- CVE-2026-42034
- CVE-2026-42035
- CVE-2026-42036
- CVE-2026-42037
- CVE-2026-42038
- CVE-2026-42039
- CVE-2026-42040
- CVE-2026-42041
- CVE-2026-42042
- CVE-2026-42043
- CVE-2026-42044
- CVE-2026-58224

# Known issues

This section describes known issues in this release of Red Hat Ceph Storage.

### Ceph Object Gateway

**Server-side copy operations can corrupt objects when `rgw_max_chunk_size` exceeds `rgw_get_obj_max_req_size`**

When `rgw_max_chunk_size` is configured with a value greater than `rgw_get_obj_max_req_size`, server-side copy operations might not copy some objects correctly. This issue affects objects that fit within a single chunk but are larger than the configured request size.

As a result, data beyond the first `rgw_get_obj_max_req_size` bytes can be copied incorrectly. The copy operation completes successfully, returns no error, and reports the source object's ETag even though the copied object is corrupted.

Currently, there is no workaround.

As a workaround, do not configure `rgw_max_chunk_size` with a value greater than `rgw_get_obj_max_req_size`. Changing these settings from their default values is not recommended.

If objects were previously copied while this configuration was in use, verify the integrity of the copied objects.

(IBMCEPH-20062)

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